



K2 AeroSim

Software Walkthrough — every workspace, every control

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K2 AeroSim is an integrated, multi-physics platform for high-power and experimental rockets: airframe design, propulsion, CFD, structural FEM, flight dynamics, avionics, 6-DOF simulation and recovery — all sharing one live rocket model. This manual walks every workspace left-to-right across the tab bar and documents each button, input box, dropdown and checkbox.

Recommended order

Design → Propulsion → (CFD for drag) → **Simulation** (fly it) → Results → Structures & Dynamics (safety) → Monte Carlo (reliability) → Optimization.

The top toolbar (always visible)

Control	What it does
New	Start a fresh project. Opens a brand-new window so the current one is left untouched. (Ctrl+N)
Open	Load a saved .k2 project file. (Ctrl+O)
Save	Save the current project to its .k2 file. (Ctrl+S)
Save As	Save the project to a new file / name. (Ctrl+Shift+S)
Import .ork	Import an OpenRocket design (.ork). Components, positions and materials are translated into the K2 component tree. (Ctrl+I)
Reset View	Recenter the 3D camera in whichever 3D view is active.
Settings	Application settings.
Run Sim (F5)	Keyboard shortcut to start the flight simulation from anywhere.
Console (bottom)	Dockable log panel. Every analysis writes progress, warnings and errors here — check it if something looks stuck. Drag it out or hide it.

1. Design

Build the airframe component-by-component and watch stability update live. Everything downstream reads this model.

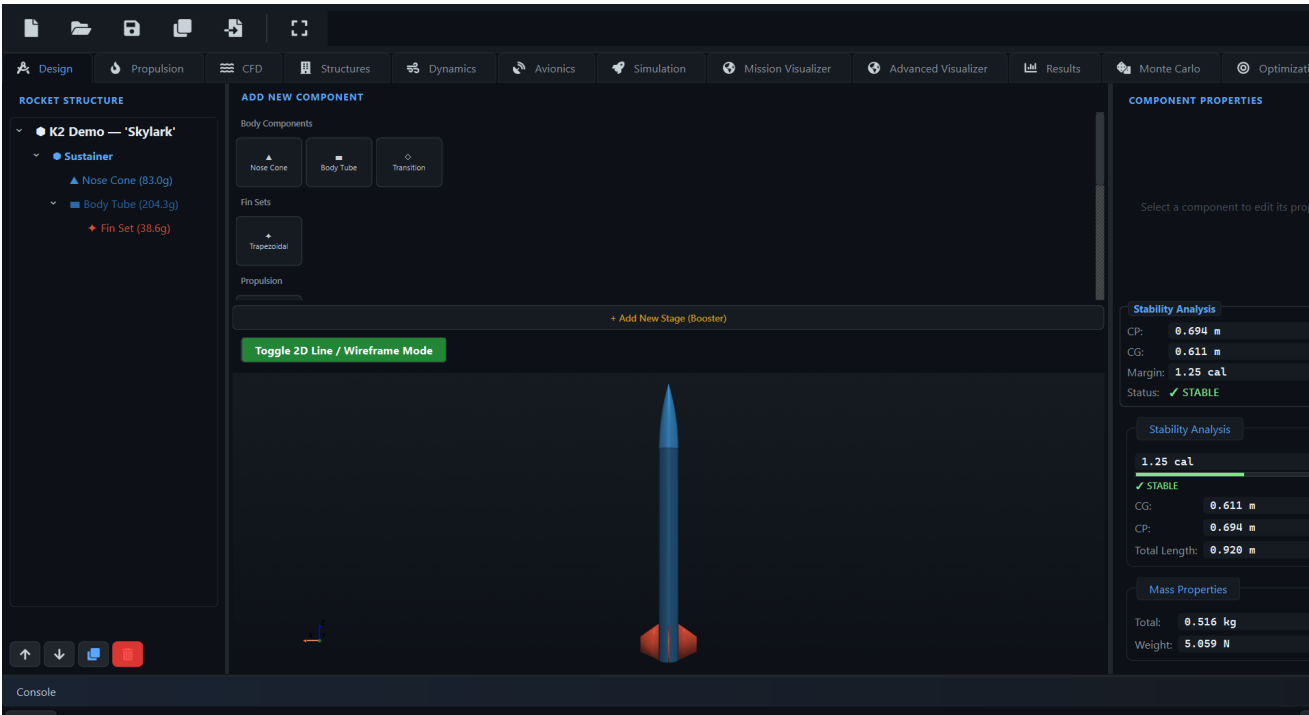


Figure 1. The Design workspace.

Rocket Structure tree (left)

Control	What it does
Component tree	The rocket as a hierarchy: Stage → body components → inner components . Click any item to select and edit it. Right-click for the same move / duplicate / delete actions as the toolbar below.
▲ Move up / ▼ Move down	Reorder the selected component within its parent. Order is nose-to-tail, so this changes where the part sits along the body.
Duplicate	Copy the selected component (with all its settings) as a sibling.
Delete	Remove the selected component.

Add New Component palette (top center)

Control	What it does
Body Components	Nose Cone , Body Tube , Transition (diameter change). These form the outer airframe and go directly under a stage.
Fin Sets	Trapezoidal fin set — count, root/tip chord, span, sweep, thickness, cross-section. Must be added onto a body tube.
Propulsion	Nozzle — external nozzle geometry.
Inner Components	Inner Tube (motor mount), Centering Ring , Bulkhead , Engine Block . Placed <i>inside</i> a body tube.
Recovery	Parachute and Shock Cord — go inside a body tube.
Mass / Attach	Mass Component (ballast / payload), Launch Lug , Rail Button .
+ Add New Stage (Booster)	Add another stage for multi-stage rockets.

3D Preview (center)

Control	What it does
Live 3D model	Rebuilds as you edit. Drag to orbit, scroll to zoom.
Toggle 2D Line / Wireframe Mode	Switch between the shaded 3D model and a flat 2D side-profile / wireframe — handy for checking proportions and station positions.

Component Properties (right)

Control	What it does
Editor fields	Edit the selected part's geometry, wall thickness, material and colour. Fields change with the component type (e.g. a nose cone exposes shape + length; a fin set exposes count, chords, sweep).

Stability Analysis (right)

Control	What it does
CP	Centre of Pressure — where aerodynamic force acts (Barrowman).
CG	Centre of Gravity — mass balance point (includes motor).
Margin	Static margin in calibers = $(CP - CG) / \text{body diameter}$.
Status	UNSTABLE / MARGINAL / ✓ STABLE / OVERSTABLE verdict from the margin.

■ **Tip:** Aim for ~1–2 cal static margin. Add a motor in Propulsion — the motor's mass shifts CG and the stability readout updates.

2. Propulsion

Pick a motor from the full [ThrustCurve.org](https://thrustcurve.org) catalogue or build a custom one.



Figure 2. The Propulsion workspace.

Motor Selection (left)

Control	What it does
Class	Filter by motor impulse class (A...O).
Diameter	Filter by motor case diameter (mm).
Fit current body	Only show motors that physically fit inside the current airframe.
Hide out-of-production	Hide discontinued motors.
Motor	The selectable list (counter shows how many pass the filters). Picking a motor applies it to the rocket and drives every analysis.
Create Custom Motor	Open the custom-motor builder: choose a grain (BATES / Tubular / End-Burner / Star), propellant and nozzle, Simulate Ballistics , then Apply to Rocket (a motor that doesn't produce thrust is flagged, not applied).

Motor Properties (left)

Control	What it does
Total Impulse / Avg Thrust / Max Thrust / Burn Time / Prop Mass	Catalogue figures for the selected motor.

Computed Performance (left)

Control	What it does
Isp / Mass Flow / Chamber P	Derived performance numbers for the motor.

Thrust Curve (center)

Control	What it does
Plot	The real measured thrust-vs-time curve (ThrustCurve.org samples). The simulation flies this curve, not a trapezoid approximation, when it is available.

■ **Tip:** The selected motor is shared with the simulation, structures, dynamics, Monte Carlo and optimization tabs.

3. CFD

SU2 aerodynamics — drag, lift, moments, pressure and flow fields.

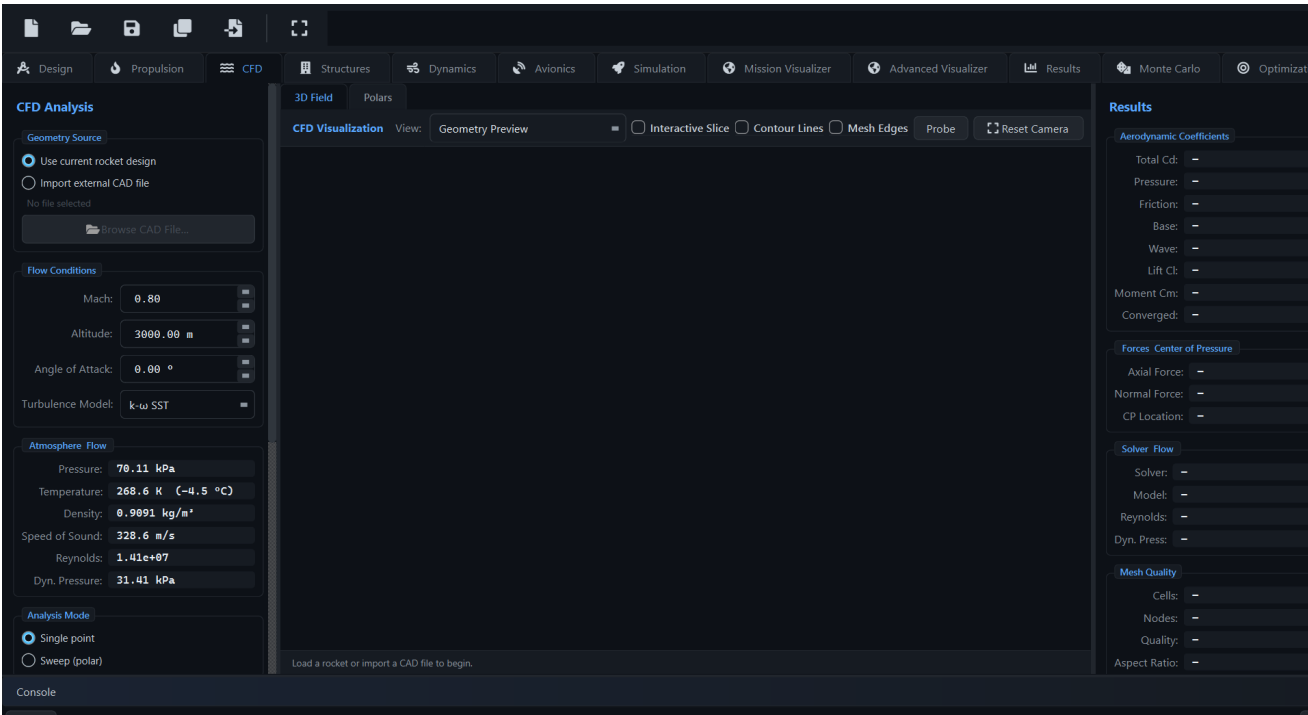


Figure 3. The CFD workspace.

Geometry Source (left)

Control	What it does
Use current rocket design	Mesh the airframe you built in Design.
Import external CAD file / Browse CAD File...	Use an external STL instead.
Export Geometry to STL	Write the current rocket as a watertight STL.

Flow Conditions (left)

Control	What it does
Mach	Freestream Mach number.
Altitude	Sets the atmosphere (pressure, temperature, density) below.
Angle of Attack	Pitch angle of the flow, degrees.
Turbulence Model	Euler / Laminar / Spalart-Allmaras / k- ω SST.

Atmosphere & Flow (left)

Control	What it does
Pressure / Temperature / Density / Speed of Sound / Reynolds / Dyn. Pressure	Read-only freestream values computed from the altitude above.

Analysis Mode (left)

Control	What it does
Single point	Solve one Mach/AoA condition.
Sweep (polar)	Solve a series over Angle of Attack or Mach (one mesh, N solves) to build a drag/lift polar — see the Polars tab.
Euler + flat-plate friction (recommended)	Fast hybrid mode: inviscid Euler pressure drag plus an analytical skin-friction estimate. Default — the tet-only RANS mesh can give untrustworthy viscous drag.

Mesh Settings (left)

Control	What it does
Refinement	Coarse (fast) ... Ultra Fine / Custom. Trades runtime for accuracy.

Run / Export (left)

Control	What it does
Run CFD Analysis	Mesh, then solve. Progress shows in the Console.
Export Results	PDF report (all contours + plots) and CSV.

Center — 3D Field tab

Control	What it does
View dropdown	Field to display: Surface Cp, Pressure/Temperature/Velocity/Mach volume slices, Streamlines, Vorticity, Q-criterion / Lambda-2, Boundary-layer Y+, surface temperature, and more (17 modes).
Interactive Slice	Drag a cutting plane through the volume field.
Contour Lines	Overlay iso-contour lines on the field.
Mesh Edges	Show the computational mesh.
Probe	Click a point to read its field value.
Reset Camera	Recenter the 3D view.

Center — Polars tab

Control	What it does
Cl / Cd / Cm vs AoA, or Cd vs Mach	Populated after a Sweep run.
Export CSV	Save the polar data.

Results (right)

Control	What it does
Aerodynamic Coefficients	Total Cd with Pressure / Friction / Base / Wave split, Lift Cl, Moment Cm, and a Converged flag.
Forces & Center of Pressure	Axial / Normal force and CP location.
Solver & Flow / Mesh Quality	Solver, turbulence model, Reynolds, cell/node counts, mesh quality and Y+ range.
Cp Distribution / Convergence Residuals / Solver Log	Pressure-coefficient plot, residual history and raw solver output.

Hand-off (right)

Control	What it does
Export VTK Results	Write the solved field for ParaView.
Map Pressure to FEM	Send the CFD surface pressure to the Structures tab as a load.

■ **Tip:** Run a Sweep to get a drag polar across AoA/Mach; the simulation can use it.

4. Structures

CalculiX FEA — stress, deflection, buckling, modes and thermal.

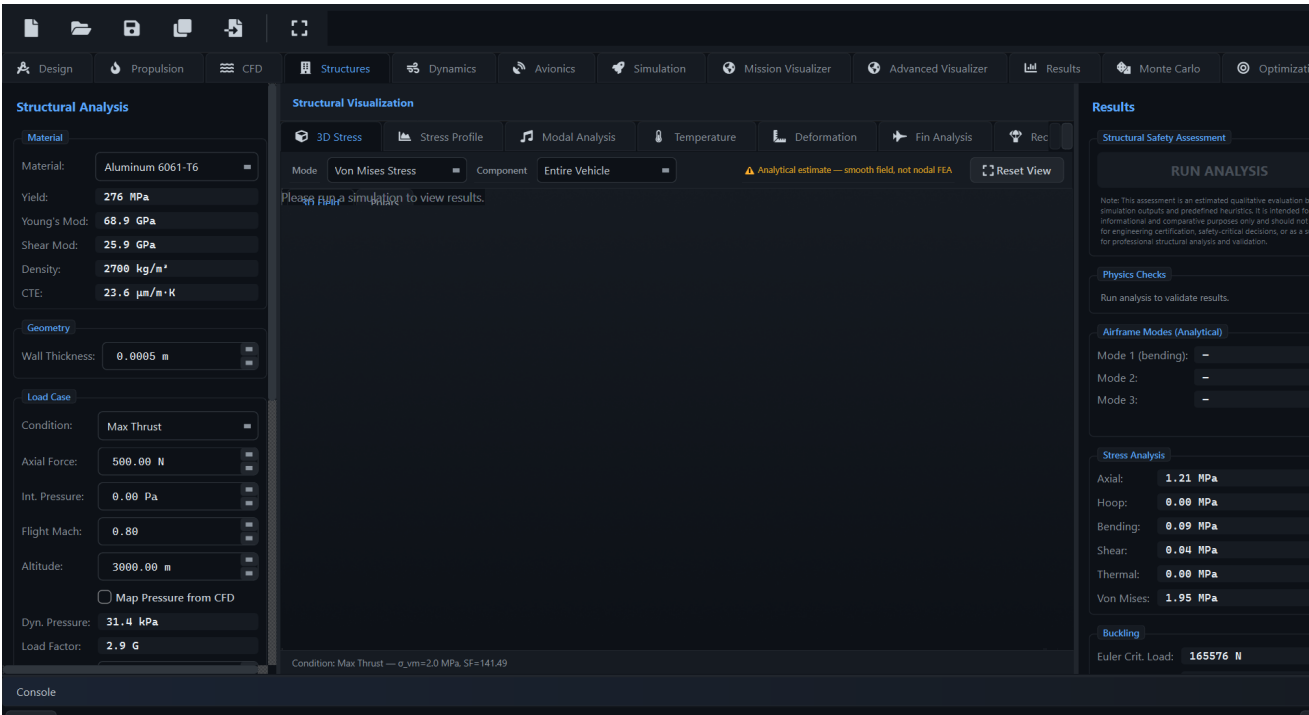


Figure 4. The Structures workspace.

Material (left)

Control	What it does
Material	Pick the airframe material (aluminium alloys, composites, ...).
Wall Thickness	Shell thickness used by the FE model.

Load Case (left)

Control	What it does
Load case	Max Thrust / Max-Q / Recovery Shock / Thermal / Custom. Each preset auto-fills a representative Mach/altitude (from your last flight when one exists).
Axial Force / Int. Pressure / Altitude / Static ΔT	The load inputs. Editable in Custom; preset in the others.
Map Pressure from CFD	Use the CFD surface-pressure field as the distributed load instead of a single axial force.

Flight Loads (left)

Control	What it does
Import flight loads	Pull the real peak loads (Max-Q etc.) from the last simulation.

FEM Settings (left)

Control	What it does
Refinement	Coarse ... Ultra Fine / Custom mesh.
Circumferential divisions / axial per-caliber	Custom mesh density controls.

Run buttons (left)

Control	What it does
Run Static Analysis	Stress + deformation under the load case.
Run Modal Analysis	Natural frequencies and mode shapes.
Run Thermal Analysis	Temperature field / thermal stress.
Export PDF Report	Full report: stress contours, deformation, fin and thermal plots.

Center result tabs

Control	What it does
3D Stress	Interactive stress contour viewer (von Mises and component stresses).
Stress Profile	Von Mises stress vs position along the airframe.
Modal / Deformation / Fin Analysis / Buckling / Temperature	Mode shapes, deflected shape, fin root stress + tip deflection, buckling modes, and thermal results.

Right column

Control	What it does
Structural Safety Assessment	Good / Marginal / Poor verdict (colour-coded) with safety factor.
Physics Checks / Airframe Modes / Stress Analysis / Buckling / Resonance / Fin Flutter / Thermal	Per-phenomenon numeric breakdown.

■ **Tip:** The smooth 3D stress map is an analytical estimate; the reported safety factor is FE-backed.

5. Dynamics

Aeroelasticity — flutter, divergence, vibration and the flight envelope.

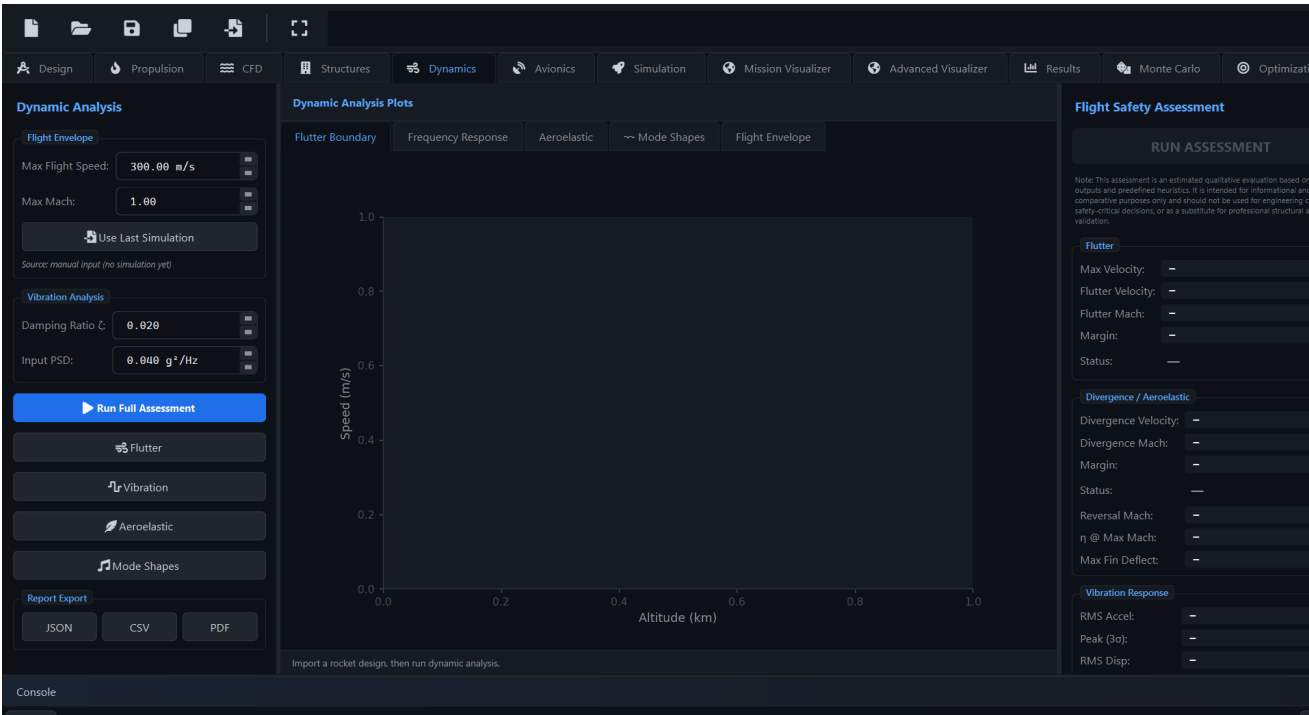


Figure 5. The Dynamics workspace.

Flight Loads (Max-Q) (left)

Control	What it does
Max-Q loads	Pulled from the trajectory — the worst-case dynamic pressure point.

Run (left)

Control	What it does
Run Full Assessment	Runs Flutter, Divergence/Aeroelastic, Vibration Response and Flight Envelope together.

Results panels

Control	What it does
Flutter	Fin flutter speed/margin vs flight Mach.
Divergence / Aeroelastic	Static aeroelastic divergence margin.
Vibration Response	Forced-response / resonance behaviour.
Flight Envelope	Safe region vs the flight profile.
Consistency Checks	Cross-checks between the phenomena and the flight data.
Flight Safety Assessment	Good / Marginal / Poor banner (colour-coded).

Report Export (left)

Control	What it does
JSON / CSV / PDF	Export the dynamics results.

■ **Tip:** Check the fin flutter margin against your maximum flight Mach.

6. Avionics

Flight-computer and sensor modelling.

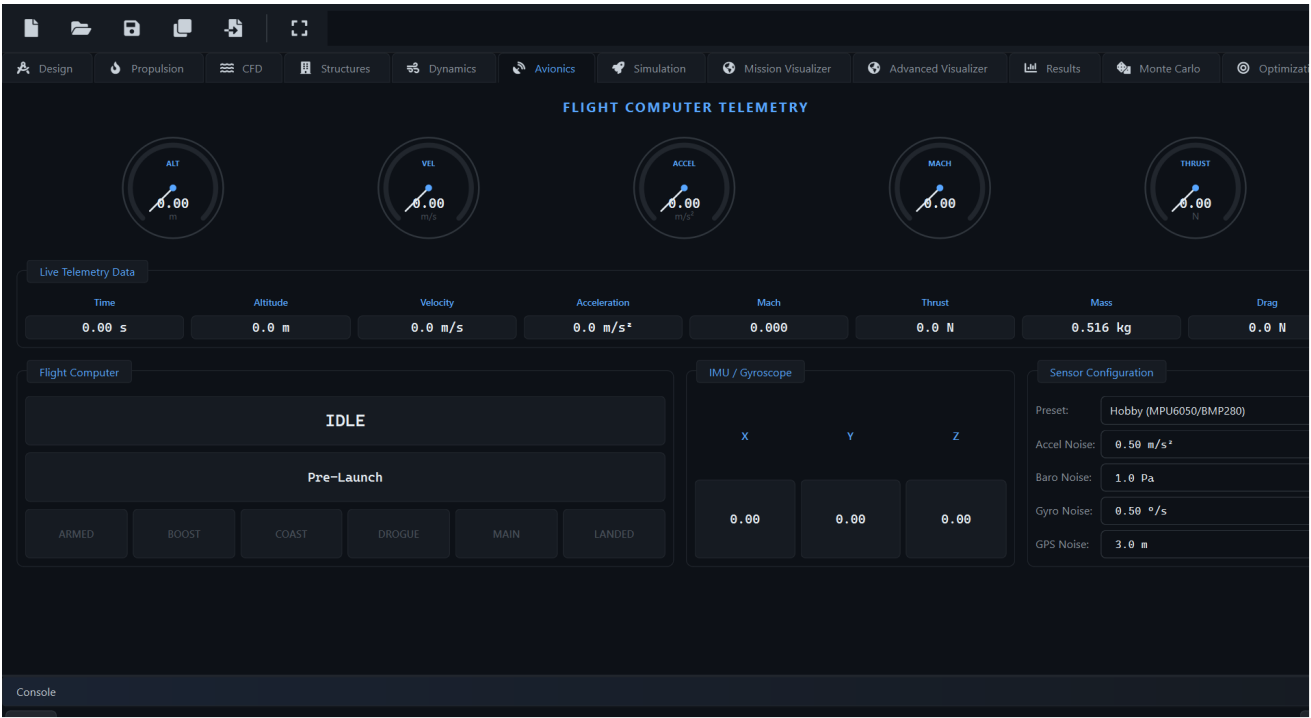


Figure 6. The Avionics workspace.

Sensor Configuration (left)

Control	What it does
IMU / Gyroscope	Inertial sensor noise / bias model.
Barometer / Sensor settings	Pressure-sensor model used for apogee detection.

Flight Computer

Control	What it does
Flight Computer settings	The simulated flight-computer / state machine.

Live Telemetry Data

Control	What it does
Telemetry readouts	Live values streamed from the flight computer during a run.

7. Simulation

Mission Control — fly the rocket with the 6-DOF solver.

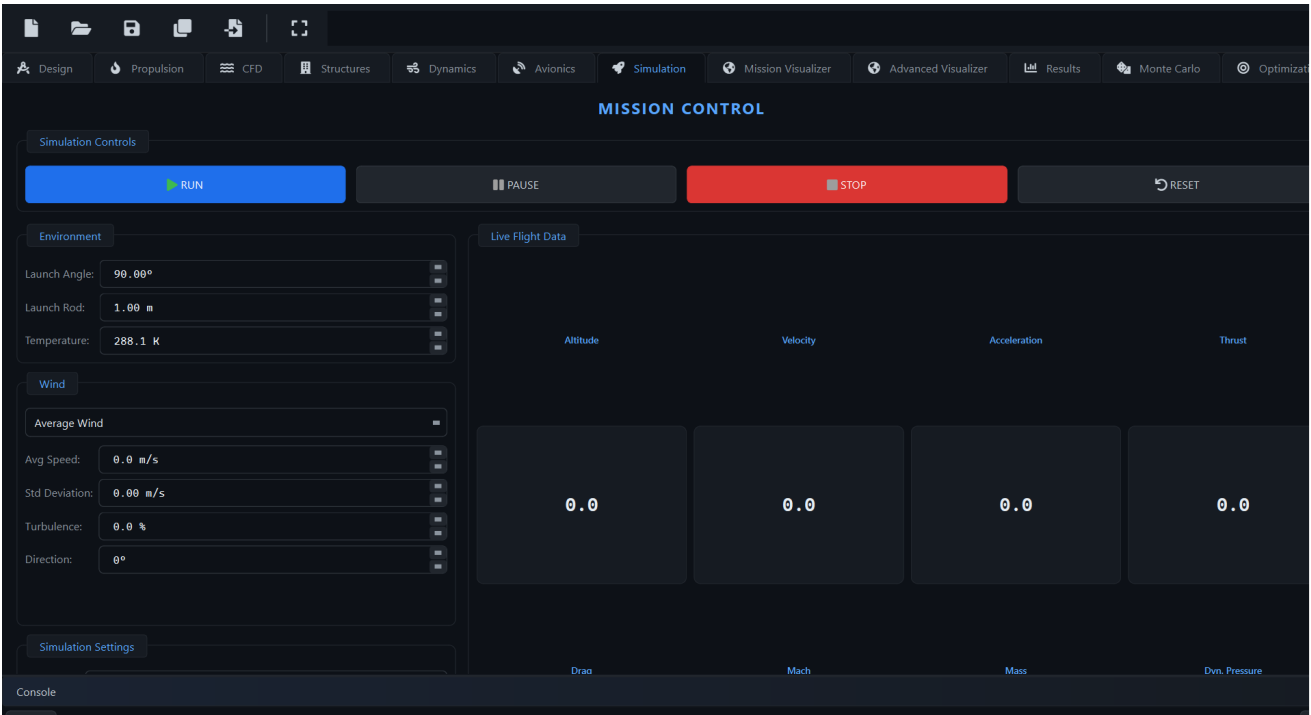


Figure 7. The Simulation workspace.

Simulation Controls (top)

Control	What it does
RUN	Start the flight. A dialog lets you pick the view: stay on the readouts, or jump to a 3D visualizer.
PAUSE	Pause / resume the running simulation.
STOP	Abort the run.
RESET	Clear the simulation state and readouts.

Environment (left)

Control	What it does
Launch Angle	Rail tilt from vertical (90° = straight up).
Launch Rod	Rail/rod length — affects off-rod velocity and weathercocking.
Temperature	Ground-level air temperature (sets the atmosphere).

Wind (left)

Control	What it does
Wind mode	Average Wind (single) or a multi-level profile.
Avg Speed / Std Deviation / Turbulence / Direction	Wind parameters.
Add Layer / Delete	Build a multi-level wind profile (altitude bands).

Simulation Settings (left)

Control	What it does
Integrator	Numerical integrator for the 6-DOF solver.
Time step	Solver step size — smaller = more accurate, slower.

Recovery System (left)

Control	What it does
Drogue / Main + deploy altitude	Parachute deployment configuration.

Live Flight Data + Flight Phase Timeline (right)

Control	What it does
Readouts	Altitude, velocity, acceleration, thrust, drag, Mach, mass, dyn. pressure — update live during the flight.
Flight Phase Timeline	Lights up Liftoff → Burnout → Apogee → Drogue → Main → Touchdown as events occur.

■ **Tip:** Run a flight first — Results, Structures, Dynamics and Monte Carlo all read it.

8. Mission Visualizer

Live 3D flight with telemetry, event flags and recovery status.

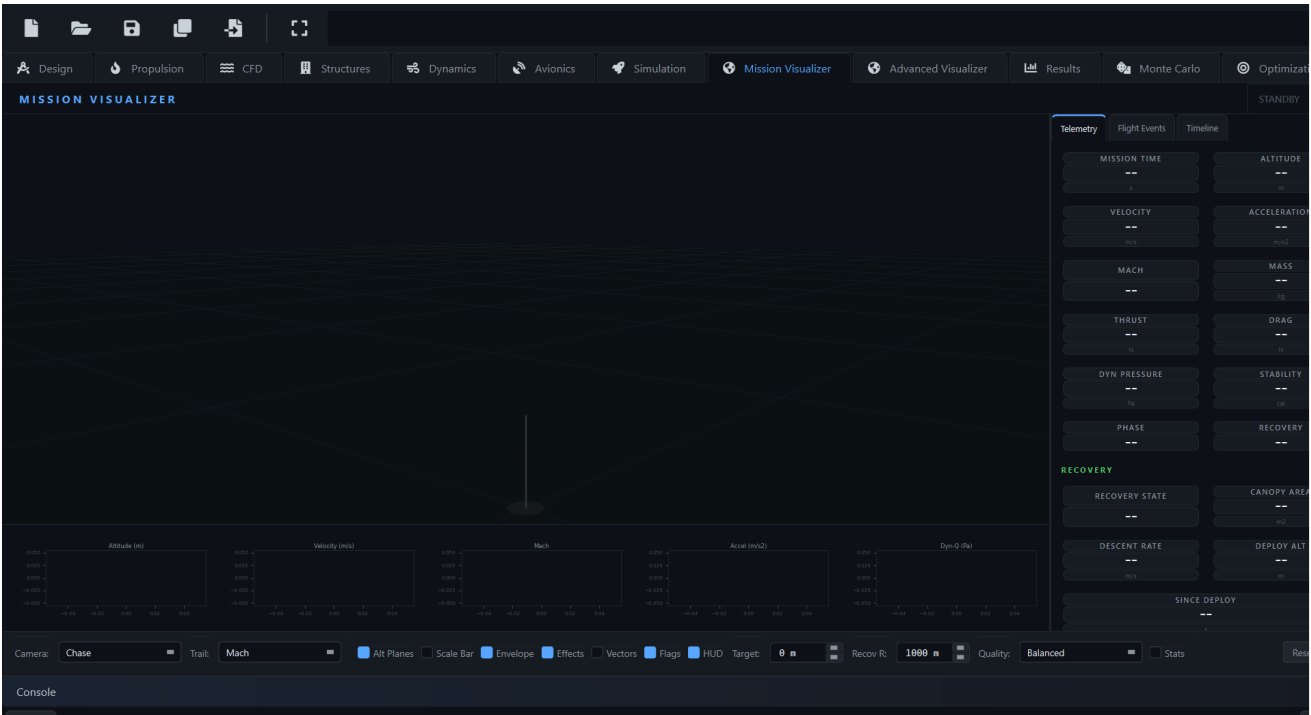


Figure 8. The Mission Visualizer workspace.

3D scene (center)

Control	What it does
Live 3D flight	Watch the rocket fly on a launch pad with a ground grid and trail.

Playback bar (bottom)

Control	What it does
Play / Pause / Restart	Control the replay.
Reset Camera	Frame the whole scene.
Camera	Camera mode (e.g. Chase).
Trail	Colour the flight trail by a channel (Mach, velocity, ...).
Alt Planes / Scale Bar / Envelope / Effects / Vectors / Flags / HUD	Toggle scene overlays.
Target / Recov R / Quality / Stats	Target altitude ring, recovery radius, render quality and an FPS/stats overlay.

Telemetry panel (right)

Control	What it does
Telemetry / Flight Events / Timeline tabs	Mission time, altitude, velocity, Mach, mass, thrust, drag, dyn. pressure, stability, phase, and the recovery block (state, descent rate, deploy altitude, canopy area).

9. Advanced Visualizer

Webcast-style cinematic replay with a SpaceX-style HUD (three.js / WebEngine).

[screenshot slot — drop docs/shots/08_advanced_visualizer.png here]

Figure 9. The Advanced Visualizer workspace.

Cinematic view

Control	What it does
Real-time 3D replay	Follow-cameras and a broadcast HUD.
Mission timeline ribbon	Liftoff, Max-Q, Burnout, Apogee, Drogue, Main, Touchdown — each shows the altitude it was reached at.
Replay scrubber + speed	Scrub the flight; LIVE jumps back to the latest run.

■ **Tip:** Needs QtWebEngine installed; otherwise the tab shows an install hint.

10. Results

Full flight history across every channel.

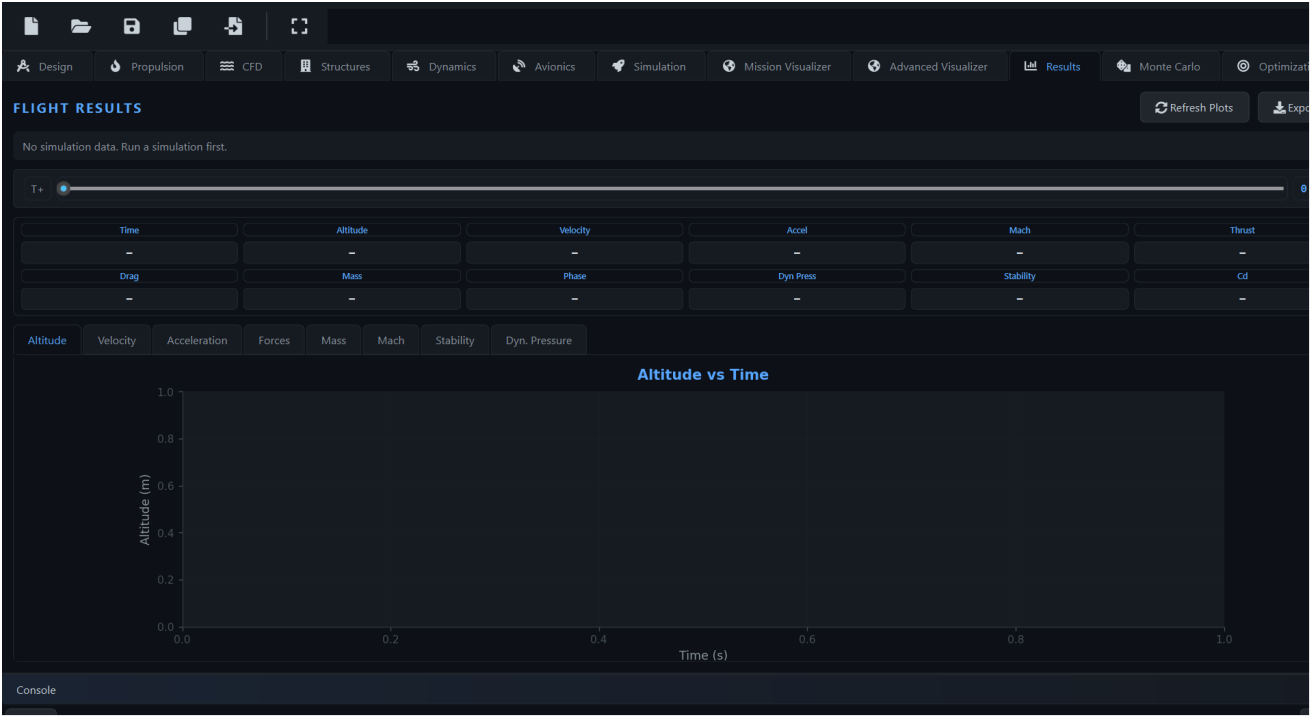


Figure 10. The Results workspace.

Plots (center)

Control	What it does
Channel plots	Scrub the entire flight: altitude, velocity, Mach, acceleration, dynamic pressure, stability, and more.

Controls

Control	What it does
Refresh Plots	Re-read the last simulation run.
Export CSV	Save the full flight data to CSV.

11. Monte Carlo

Uncertainty and reliability over many randomized flights.

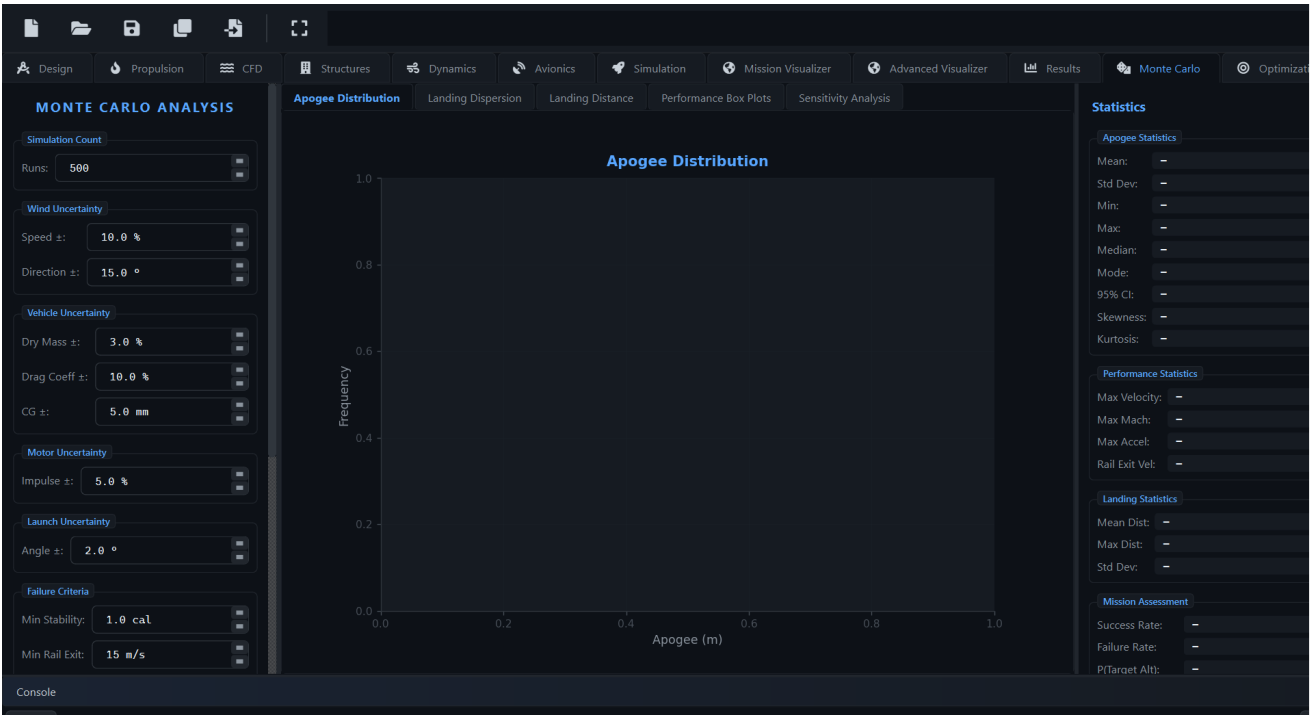


Figure 11. The Monte Carlo workspace.

Inputs (left)

Control	What it does
Launch Uncertainty	Spread on launch angle / conditions.
Motor Uncertainty	Spread on motor performance.
Simulation Count	Number of randomized runs (more = tighter stats, longer runtime).
Target	Target apogee to score against.
Failure Criteria	What counts as a failed flight.

Run / Export (left)

Control	What it does
RUN ANALYSIS	Run the batch.
CANCEL	Stop a running batch.
EXPORT CSV	Per-run data.
EXPORT PDF	Stats + distribution plots.

Results

Control	What it does
Apogee / Landing / Performance Statistics	Mean, spread and percentiles.
Reliability Analysis / Failure Breakdown / Mission Assessment	Success probability, failure causes, overall verdict.
Best Case / Worst Case	Jump to the extreme runs.

■ **Tip:** More runs → tighter statistics, longer runtime.

12. Optimization

Multidisciplinary design optimization (MDO).

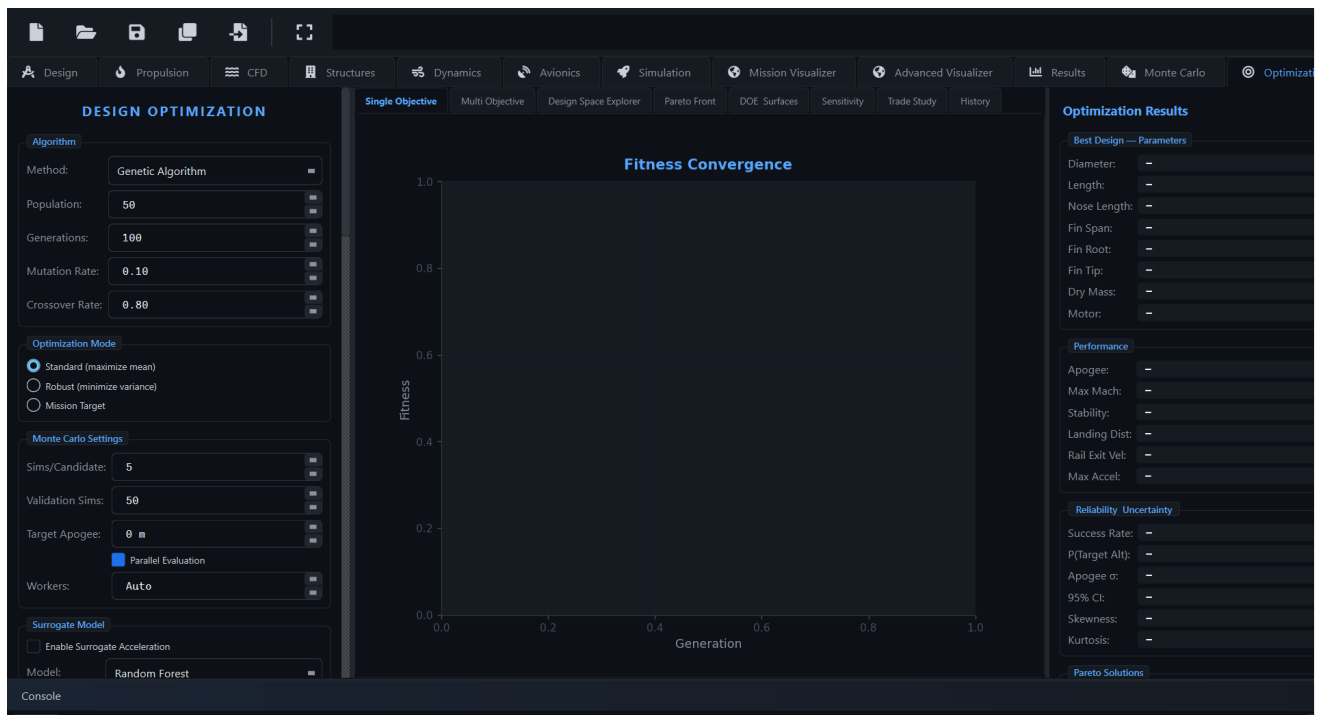


Figure 12. The Optimization workspace.

Setup (left)

Control	What it does
Algorithm	GA, NSGA-II (multi-objective), Differential Evolution or Particle Swarm.
Optimization Mode	Standard, Mission Target (hit an apogee), or Robust.
Target Apogee	Goal altitude for Mission Target mode (0 = auto).
Surrogate Model	Optional surrogate to speed up evaluations.
Monte Carlo Settings	Sampling used for robust / reliability objectives.
Design Variables	Enable variables and set their bounds (Select All / Deselect All).
Objectives	Pick objectives and weight/direction; statistical objectives expose a mode (mean / std / worst / reliability / p5).
Constraints	e.g. Stability > 1.2 and < 3.0.

Actions (left)

Control	What it does
START OPTIMIZATION	Run the optimizer.
CANCEL	Stop the run.
CSV / JSON / PDF	Export results.

Result tabs (center)

Control	What it does
Single / Multi Objective	Convergence and objective space.
Design Space Explorer	Scatter the sampled designs; colour by Fitness / Feasibility / Apogee / Stability / Mach.
Pareto Front	Trade-off curve for multi-objective runs; click a point to inspect it.
DOE & Surfaces	Latin Hypercube / Full Factorial / Taguchi designs and response surfaces (Run DOE).
Sensitivity	Sobol indices / PRCC / Morris screening (Analyze).
Trade Study	Compare named configurations on a radar plot (Add Current Design / Add Best Optimized / Compare / Clear).
History	Every evaluated design.

Summary (right)

Control	What it does
Best Design — Parameters	Winning design variables.
Performance / Reliability & Uncertainty	Apogee, P(Target Alt), etc.
Improvement Over Baseline / Constraint Status / Optimization Statistics	How much better, and whether constraints are met.
Pareto Solutions	Shortcuts: Best Apogee / Best Reliability / Best Mass Efficiency / Best Balanced.

■ **Tip:** Set a Target Apogee (Mission Target mode) to make the optimizer hit a number rather than just maximize altitude.

Saving & exporting

Only the design and component tree are stored in a **.k2** project — all analyses recompute from the design. To keep CFD, Structures, Dynamics, Monte Carlo or Optimization results, use that tab's **Export** (PDF / CSV / JSON). Free and open source under GPL-3.0.